



First Principles Driven Hybrid Model for Atmospheric Crude Distillation Tower Efficiency

Starter File: Atmospheric_Crude_Tower_Starter.hsc
CDU Plant Data_version500_assay.xlsm
Solution file: Atmospheric_Crude_Tower_Deployed.hsc
Version: V12.1

Atmospheric Crude Distillation Tower is the most important unit of a refinery, hence the importance of understanding the impact of plant conditions on the column to accurately represent field operations and optimize the production. The objective of this example is to build a predictive model to estimate the efficiency of an Atmospheric Distillation Tower using Plant Data with First Principles-Driven Hybrid Modeling Technology in Aspen HYSYS.

In this document, we highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model

General Workflow

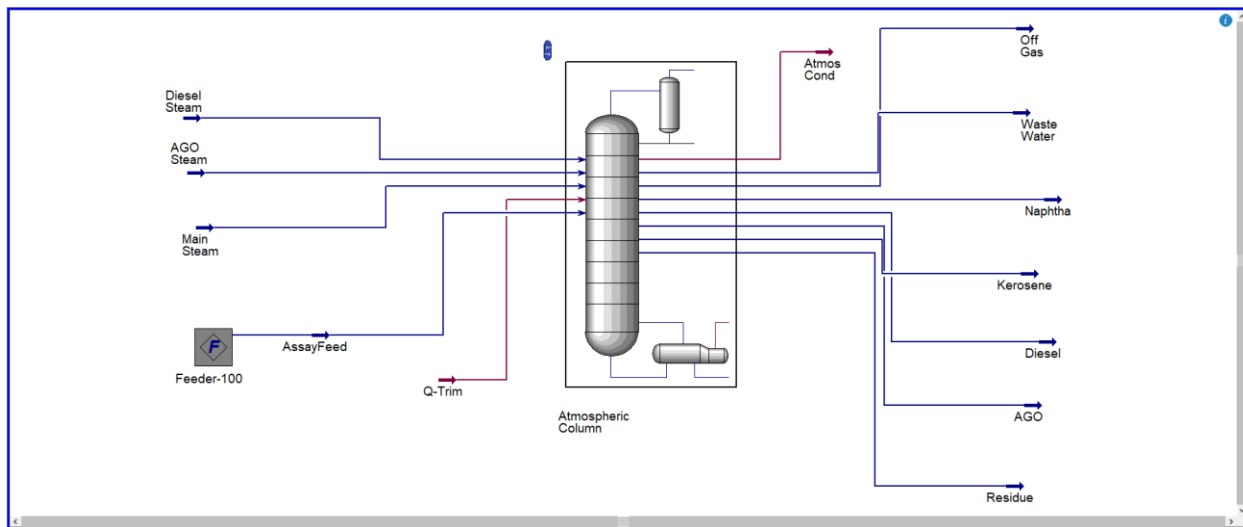


Atmospheric Crude Distillation Tower Efficiency

1. Start by opening **Atmospheric_Crude_Tower_Starter.hsc**

Note: This example library was developed in V12.1

The starter file includes the First Principles model in HYSYS representing a generic Atmospheric Crude Distillation Tower modelled as a rigorous column in which major products and main control parameters are known. The model is fully solved.



In this model, tray efficiencies are grouped. You can open the Group Definition to understand how the trays are distributed across the 8 groups given. The intention is to maintain Group-0 efficiency as 1. The rest of the groups (Group-1 to Group-7) are considered unknown variables, and the current displayed values are estimates which are constant. Our AI model will help us to properly estimate these efficiency values.

Column: Atmospheric Column / COL1 Fluid Pkg: Basis-1 / PengRobinson

Design Parameters Side Ops Internals Rating Worksheet Performance Flowsheet Reactions Dynamics

Parameters

Profiles
Estimates
Efficiencies
Solver
2/3 Phase
Fluid Pkgs

Stage Efficiencies

Efficiency Type
☒ Overall
☐ Component

Efficiency Values
☒ Grouped
☐ User Specified

Eff. Multi-Spec
Specify ---->

	Stage Efficiency
Condenser	1.000
1_Main TS	1.000
2_Main TS	1.000
3_Main TS	0.4000
4_Main TS	0.4000
5_Main TS	0.4000
6_Main TS	0.4000
7_Main TS	0.4000
8_Main TS	0.4000
9_Main TS	0.4000
10_Main TS	0.4000
11_Main TS	0.4000
12_Main TS	0.4000
13_Main TS	0.4000
14_Main TS	0.4000
15_Main TS	0.4000
16_Main TS	0.4000
17_Main TS	0.4000
18_Main TS	0.4000
19_Main TS	0.4000
20_Main TS	0.4000
21_Main TS	0.4000
22_Main TS	0.4000
23_Main TS	0.4000
24_Main TS	0.4000
25_Main TS	0.4000
26_Main TS	0.4000
27_Main TS	0.4000
28_Main TS	1.000
29_Main TS	1.000
1_Kero_SS	1.000
2_Kero_SS	1.000
3_Kero_SS	1.000

Efficiency Tuning By Group

Group Definition ☐ Show Spec Summary

Group Efficiency Values

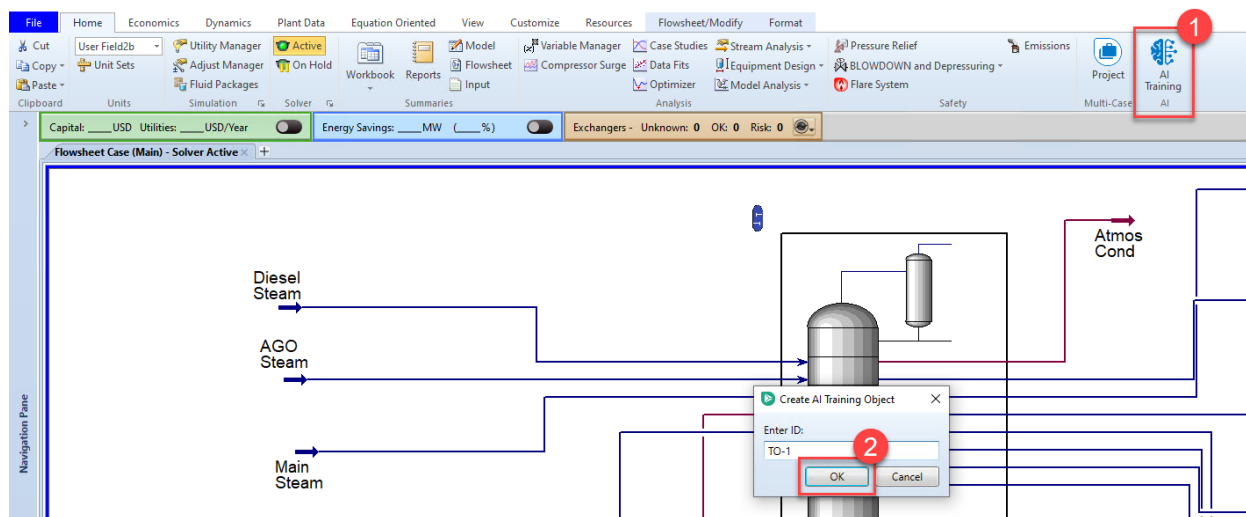
Group Name	Value
Group-0	1.000
Group-1	0.4000
Group-2	0.4000
Group-3	0.4000
Group-4	0.4000
Group-5	0.4000
Group-6	0.4000
Group-7	0.4000

Delete Column Environment... Run Reset Converged

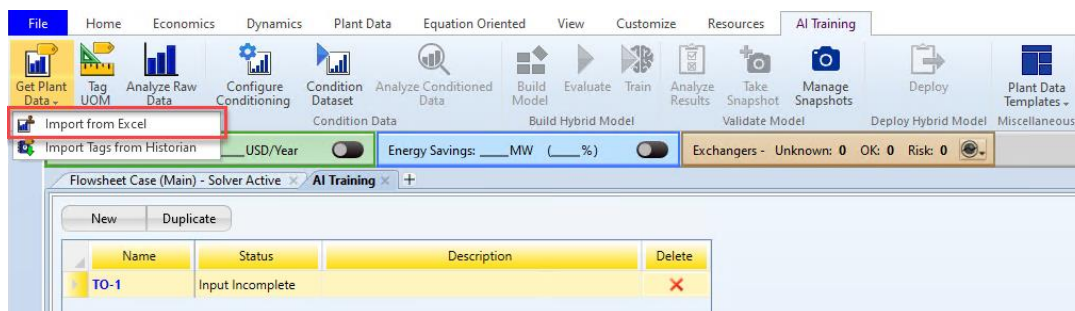


Step 1 - Import Data

2. Click the **AI Training** icon, located at **Home tab | AI**. Click OK for the AI Training Object manager; the default name for our model TO-1.

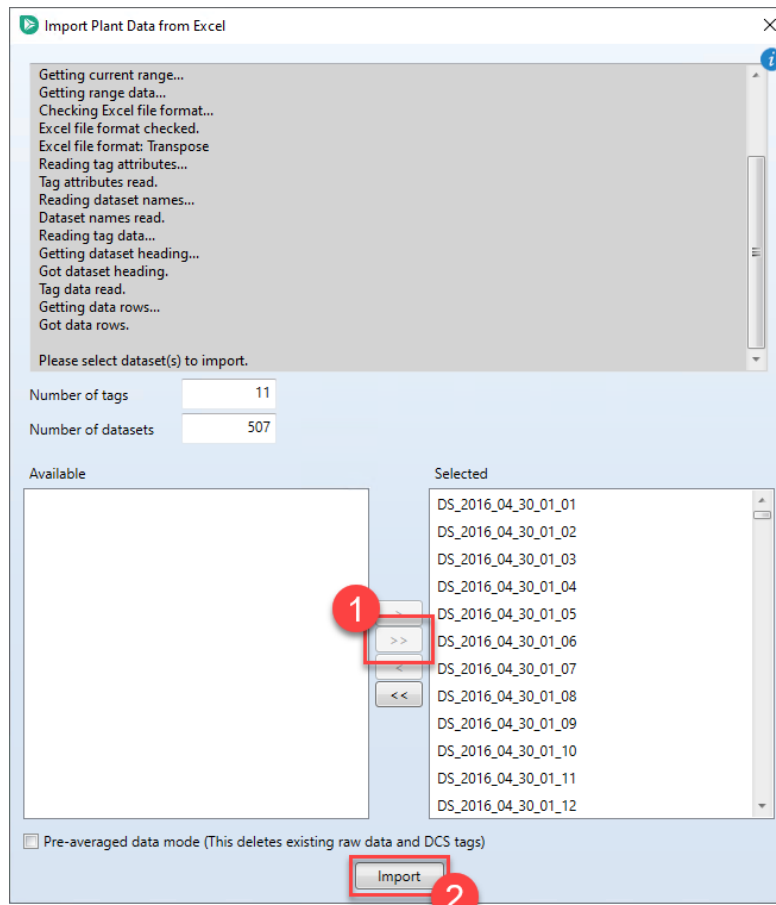


3. The AI Training tab will enable. Click the **Get Plant Data** button and from the options select **Import from Excel**, browse for the **CDU Plant Data_version500_assay.xlsm** file and click open.



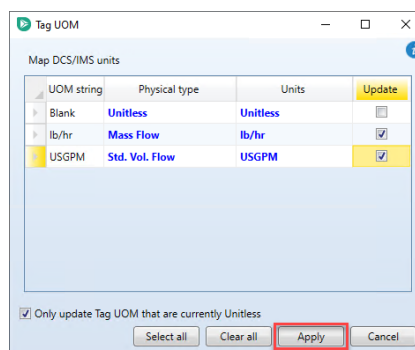
Note: Remember that the organization of this tab starts on the left side uploading the data and finishes with the deployment of your AI model at the right end.

4. The Import Plant Data manager will display, and information will be loaded showing 11 tags and 507 data sets. Select them all, click the >> button to transfer all 507 datasets; then click **Import**.



*Note: Once all data is imported the Import manager will display the message **Plant Data Successfully Imported**.*

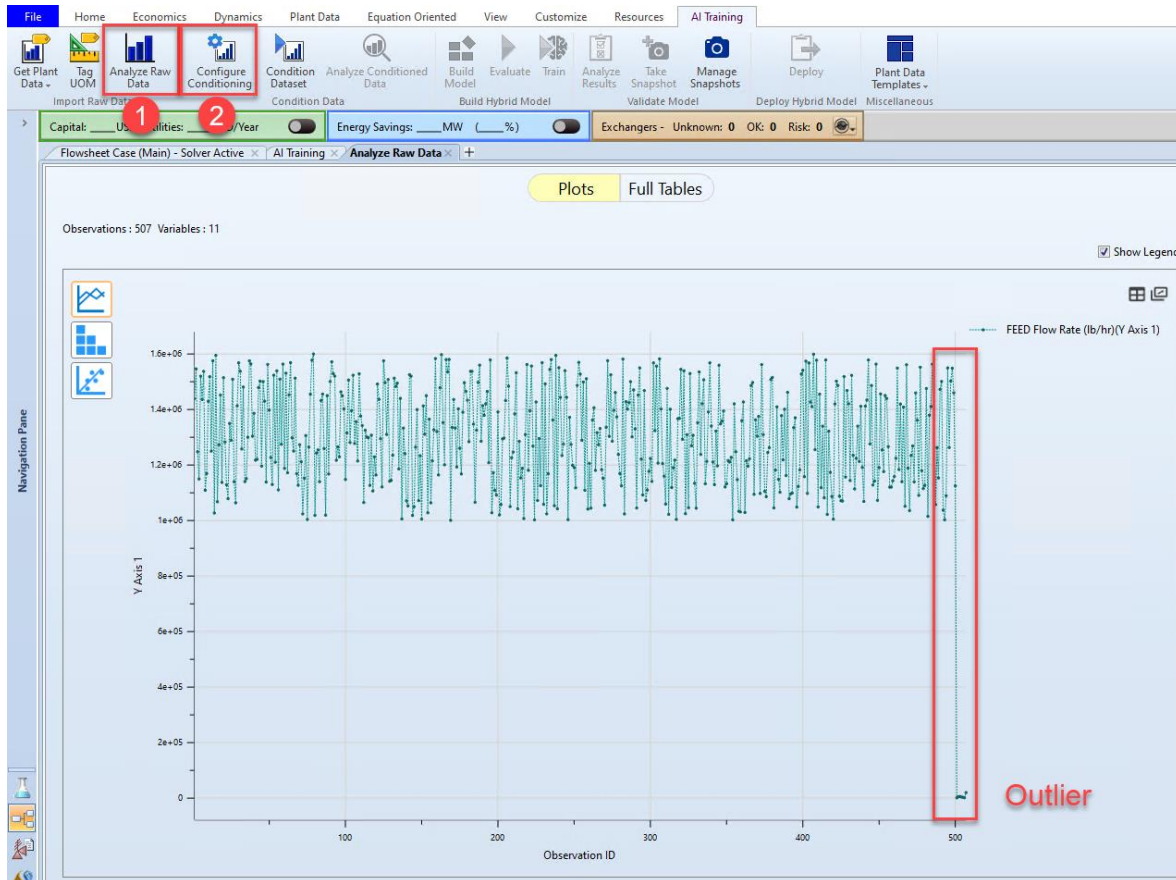
5. Close the Import Plant Data from Excel window, and Map the DCS/IMS units, use lb/hr for mass flow, and USGPM for Std. Vol. Flow, check the corresponding Update checkboxes and then Apply.



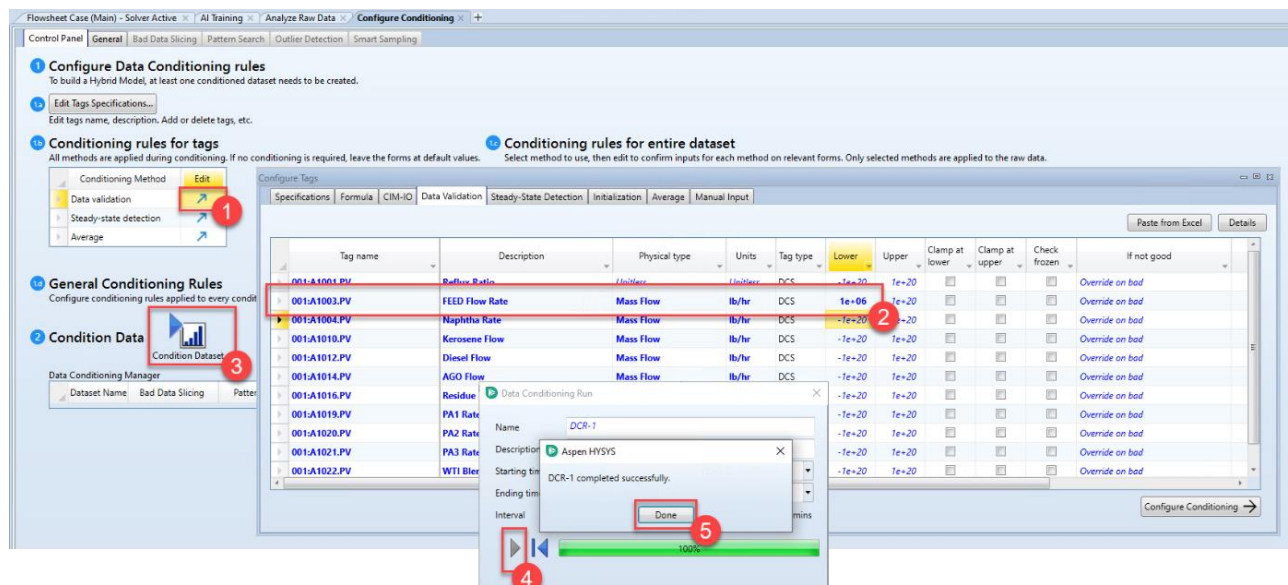


Step 2 – Condition Data

6. Use the different plots and tables given in **Analyze Raw Data** section to explore the data recently imported. FEED Flow Rate variable has an outlier that should be removed. Click the **Configure Conditioning** in the AI Training ribbon section.



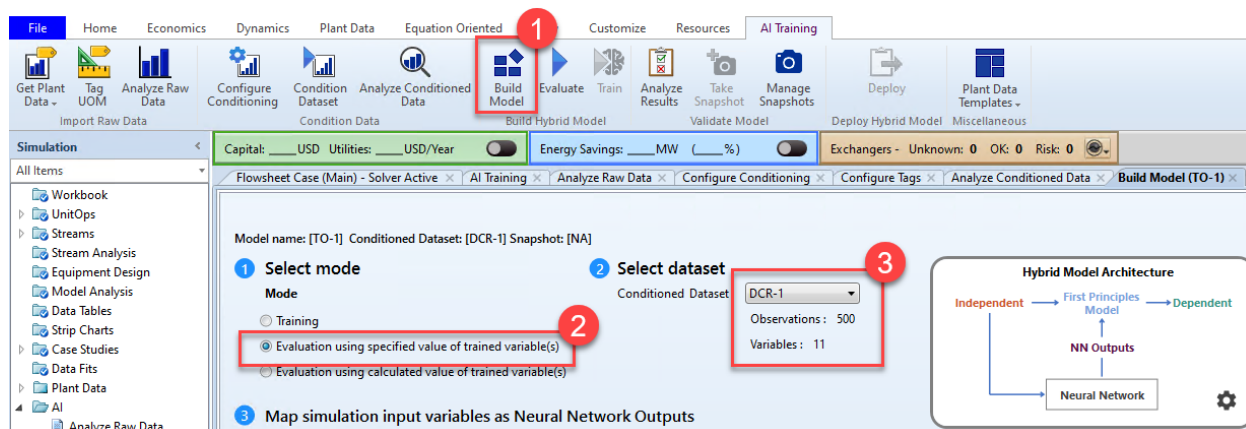
7. Use the Data validation conditioning method and click the edit arrow to open the Configure Tags Window. Locate FEED Flow Rate variable and define a lower bound of $1\text{E}+06$ lb/hr. Go back to the previous view (Configure Conditioning) and click **Condition Dataset** button; the Data Conditioning Run manager will appear, click Run then Done.



Note: Use Analyze Conditioned Data to observe in graphical and tabular format the changes applied to your data after conditioning step.

Step 3 – Evaluate Model

- Click the **Build Model** button in the AI Training ribbon and select the mode: Evaluation using specified value of trained variable(s). We want to check if the current model represents the data adequately and if there is a need to build a hybrid model. DCR-1 will be set as the conditioned set with 500 observations and 11 variables.



- Map the simulation input variables. For input variables as Neural Network Outputs you can browse or copy and paste the 7 Group Efficiency Values from the Atmospheric Crude Tower (Parameters | Efficiencies | Group Efficiency Values). Define a Lower Limit of 0.4, Upper Limit of 1 and Initial value of 0.4 for all the trained variables.

**3 Map simulation input variables as Neural Network Outputs**

New Copy Paste

Trained Variable	Description	Lower Limit	Upper Limit	Initial	Units	Delete
FlowSheetObjects.UnitOps.Atmospheric Column.Group Efficiency Values.Group-1	Efficiency G1	0.4	1	0.4		✗
FlowSheetObjects.UnitOps.Atmospheric Column.Group Efficiency Values.Group-2	Efficiency G2	0.4	1	0.4		✗
FlowSheetObjects.UnitOps.Atmospheric Column.Group Efficiency Values.Group-3	Efficiency G3	0.4	1	0.4		✗

10. Map the tags as independent or dependent variables, you can browse for them or copy and paste as follows:

Description	Stream/Unit Operation	Variable Name	Dependent/Independent
Reflux Ratio	Atmospheric Column	Reflux Ratio (Column Spec Value)	Independent
FEED Flow Rate	AssayFeed	Mass Flow	Independent
Naphtha Rate	Naphtha	Mass Flow	Dependent
Kerosene Flow	Kerosene	Mass Flow	Dependent
Diesel Flow	Diesel	Mass Flow	Dependent
AGO Flow	AGO	Mass Flow	Dependent
Residue Flow	Residue	Mass Flow	Dependent
PA1 Rate	Atmospheric Column	PA_1_Rate(Pa) (Column Spec Value)	Independent
PA2 Rate	Atmospheric Column	PA_2_Rate(Pa) (Column Spec Value)	Independent
PA3 Rate	Atmospheric Column	PA_3_Rate(Pa) (Column Spec Value)	Independent
WTI Blend	Feeder-100	Wti Domestic Sweet-2016 Flow Ratio	Independent

4 Map Tags to simulation variables or to user-defined custom variables

Assign each Tag as Independent or Dependent (simulation inputs = Independent, simulation outputs = Dependent)

Include in Model	Tag	Description	Units	Variable Type	Variable Name	Independent	Dependent
☒	001:A1001.PV	Reflux Ratio		Simulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Column Spec.Reflux Ratio.Spec Value	☒	☐
☒	001:A1003.PV	FEED Flow Rate	lb/hr	Simulation	FlowSheetObjects.Streams.AssayFeed.Mass Flow	☒	☐
☒	001:A1004.PV	Naphtha Rate	lb/hr	Simulation	FlowSheetObjects.Streams.Naphtha.Mass Flow	☐	☒
☒	001:A1010.PV	Kerosene Flow	lb/hr	Simulation	FlowSheetObjects.Streams.Kerosene.Mass Flow	☐	☒
☒	001:A1012.PV	Diesel Flow	lb/hr	Simulation	FlowSheetObjects.Streams.Diesel.Mass Flow	☐	☒
☒	001:A1014.PV	AGO Flow	lb/hr	Simulation	FlowSheetObjects.Streams.AGO.Mass Flow	☐	☒
☒	001:A1016.PV	Residue Flow	lb/hr	Simulation	FlowSheetObjects.Streams.Residue.Mass Flow	☐	☒
☒	001:A1019.PV	PA1 Rate	USGPM	Simulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Column Spec."PA_1_Rate(Pa)".Spec Value	☒	☐
☒	001:A1020.PV	PA2 Rate	USGPM	Simulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Column Spec."PA_2_Rate(Pa)".Spec Value	☒	☐
☒	001:A1021.PV	PA3 Rate	USGPM	Simulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Column Spec."PA_3_Rate(Pa)".Spec Value	☒	☐
☒	001:A1022.PV	WTI Blend		Simulation	FlowSheetObjects.UnitOps.Feeder-100.Flow Ratio.Wti Domestic Sweet-2016	☒	☐

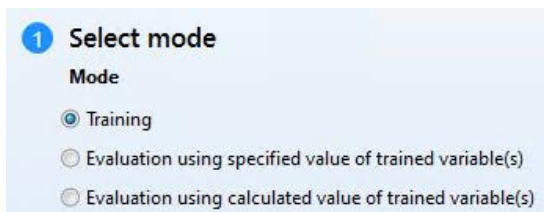
11. Click **Evaluate** from the ribbon and observe the results after the run. This evaluation demonstrates the model is not good with the current values specifically for heavier products. Hence, we want to train to improve the model and create the hybrid model.

	Naphtha Rate	Kerosene Flow	Diesel Flow	AGO Flow	Residue Flow
R ²	0.9996	0.9803	0.7067	0.9614	0.9249



Step 4 – Build, Train and Validate Model

12. Select Training mode.



13. Click **Neural Network** button under Hybrid Model Architecture manager and for Training set fraction enter 0.8, Hidden layers 2, Training Epochs 100. Set 8 Neurons for both layers and select Sigmoid for the activation. Expand the Advance options and select Adam method. Click Train.

Hybrid Model Architecture

Independent → First Principles Model → Dependent

NN Outputs

Neural Network

Neural Network Configuration

Basic

Training set fraction: 0.8 Hidden Layers: 2 Training Epochs: 100 Learning Rate: 0.001

Hidden Layer	Neurons	Activation	Activation Parameter
1	8	Selu	0.1
2	8	Selu	0.1

Output Layer

Predicted Parameter	Activation	Activation Parameter	Lower	Upper
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1
FlowSheetObjects.UnitOps.Atmospheric C...	Sigmoid	0.1	0.4	1

Initialization

☒ Cold start

☐ Reinitialize from previously converged run

Performance

☒ Use parallel computing during training

Advanced

☐ Adaptive learning rate

☒ Scale objective function

☒ Delete temporary files after training

☒ Shuffle training set on each pass

Batch Size: 16

Decay rate: 1E-05

Learning rate multiplier: 0.9

Method: Adam

Variables

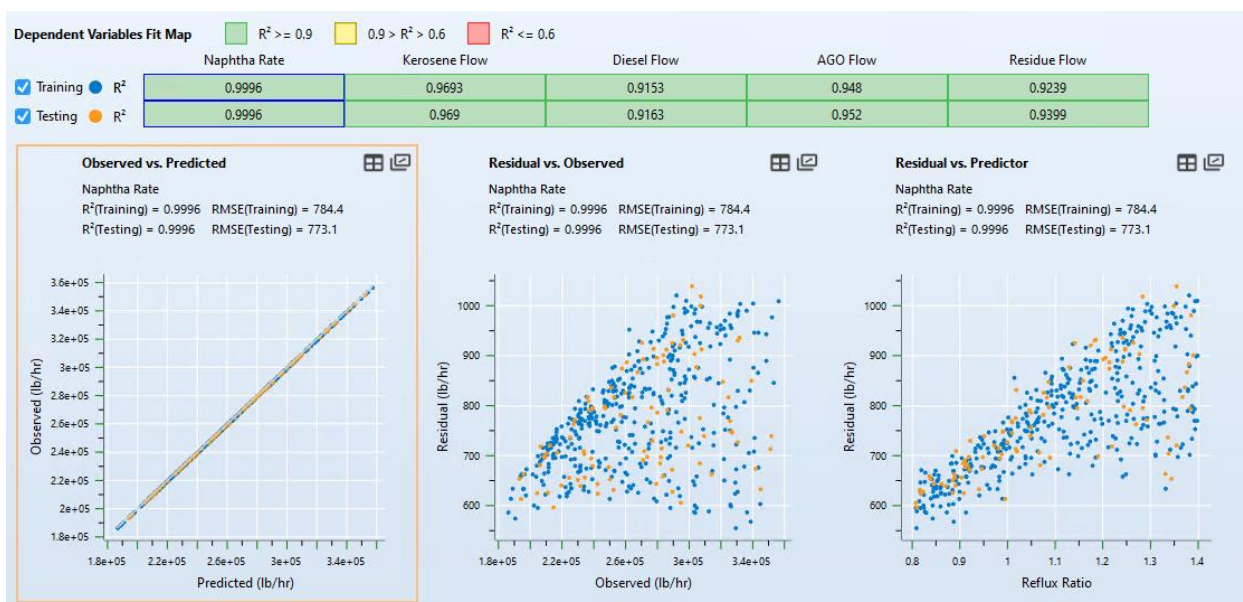
ts = Independent, simulation outputs = Dependent

Variable Type	Variable Name
mulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Colu
mulation	FlowSheetObjects.Streams.AssayFeed.Mass Flow
mulation	FlowSheetObjects.Streams.Naphtha.Mass Flow
mulation	FlowSheetObjects.Streams.Kerosene.Mass Flow
mulation	FlowSheetObjects.Streams.Diesel.Mass Flow
mulation	FlowSheetObjects.Streams.AGO.Mass Flow
mulation	FlowSheetObjects.Streams.Residue.Mass Flow
mulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Colu
mulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Colu
mulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Colu
mulation	FlowSheetObjects.UnitOps.Atmospheric Column.My SubFlowsheet.Variables.Any Colu



Note: Visit the AI Training section in the Aspen HYSYS Help Menu (F1) to learn more about the Neural Network Configuration options.

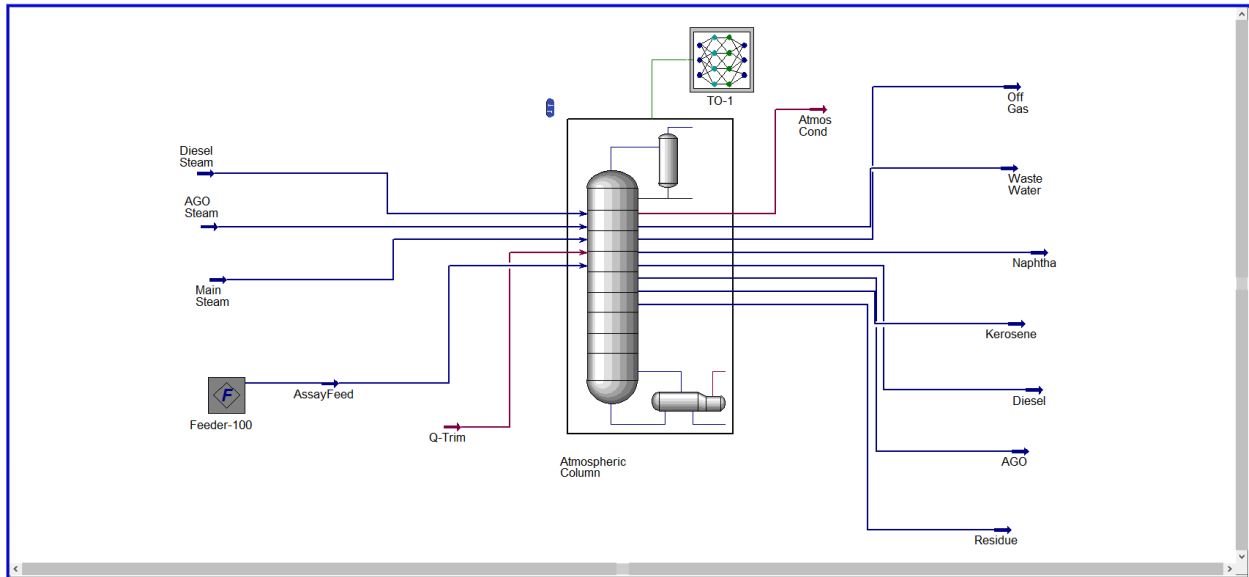
14. Once the training finalizes, observe predictions are good within reasonable ranges. Use the R² results and the different plots to compare testing vs training variables and observe the relationship between Dependent Variables and the Column Efficiencies.



Note: Feel free to experiment changing the neural network configuration and train the model. You can save a snapshot after each training to compare the results.

Step 5 – Deploy Model

15. Model is ready to be deployed. Click **Deploy** and the AI Calculator TO-1 will be attached to the Distillation Column. The Connections tab shows all the involved variables where NN Outputs correspond to the column efficiencies that are calculated by the AI Calculator. Try adjusting any input (e.g., Reflux Ratio to 0.85) and observe the efficiencies are properly estimated. These values are fully connected to the tray group efficiencies in the Atmospheric Crude Tower.



AI Calculator: TO-1

Connections Parameters About

AI Calculator Name: TO-1

Neural Network Inputs

Tag	Object	Variable Description	Value	Unit	Physical Type	Lower Limit	Upper Limit	User Description
001:A1001.PV	Atmospheric Column	Spec Value (Reflux Ratio)	0.8500	Unitless	Unitless	0.8072	1.399	Reflux Ratio
001:A1003.PV	AssayFeed	Mass Flow	1.100e+006	lb/hr	Mass Flow	1.001e+006	1.600e+006	FEED Flow Rate
001:A1019.PV	Atmospheric Column	Spec Value (PA_1_Rate(Pa))	1458	USGPM	Liq. Vol. Flow	1400	1500	PA1 Rate
001:A1020.PV	Atmospheric Column	Spec Value (PA_2_Rate(Pa))	875.0	USGPM	Liq. Vol. Flow	850.1	949.6	PA2 Rate
001:A1021.PV	Atmospheric Column	Spec Value (PA_3_Rate(Pa))	930.0	USGPM	Liq. Vol. Flow	850.4	950.0	PA3 Rate
001:A1022.PV	Feeder-100	Flow Ratio (Wt Domestic Sweet-2016)	0.3000	Unitless	Unitless	1.988e-003	0.5996	WTI Blend

Neural Network Outputs

Tag	Object	Variable Description	Value	Unit	Physical Type	Lower Limit	Upper Limit	User Description
NNOutput-1	Atmospheric Column	Group Efficiency Values (Group-1)	0.4027	Unitless	Unitless	0.4000	1.000	Efficiency G1
NNOutput-2	Atmospheric Column	Group Efficiency Values (Group-2)	0.4102	Unitless	Unitless	0.4000	1.000	Efficiency G2
NNOutput-3	Atmospheric Column	Group Efficiency Values (Group-3)	0.4680	Unitless	Unitless	0.4000	1.000	Efficiency G3
NNOutput-4	Atmospheric Column	Group Efficiency Values (Group-4)	0.4083	Unitless	Unitless	0.4000	1.000	Efficiency G4
NNOutput-5	Atmospheric Column	Group Efficiency Values (Group-5)	0.4054	Unitless	Unitless	0.4000	1.000	Efficiency G5
NNOutput-6	Atmospheric Column	Group Efficiency Values (Group-6)	0.4084	Unitless	Unitless	0.4000	1.000	Efficiency G6
NNOutput-7	Atmospheric Column	Group Efficiency Values (Group-7)	0.4059	Unitless	Unitless	0.4000	1.000	Efficiency G7

Delete [Progress Bar] Ignored

Note: Remember to save your AI Training files as HYSYS Compound files (.hscz)